\HOM_\MATHCAL{O}(J_!\MATHCAL{O}_U, -) COMMUTES WITH DIRECT SUMS FOR QUASI-COMPACT U

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one displayed equation of the same chapter and one same-chapter lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Maps out of j_!\mathcal{O}_U into a module. Stacks Project, Modules on Sites, displayed equation equation-map-lower-shriek-OU-into-module. Supplies the identification \Hom(j_!\mathcal{O}_U, \mathcal{F}) = \mathcal{F}(U).
- Sections over quasi-compact objects commute with direct sums. Stacks Project, Modules on Sites, Lemma lemma-sections-over-quasi-compact (carded locally as S11-sites-mod-sections-over-quasi-compact). Supplies that \mathcal{F} \mapsto \mathcal{F}(U) commutes with direct sums for quasi-compact U.

Source reference. Original equation 0G1V: sources/stacks-chow/tags.txt, source lines 20035–20035. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0935: sources/stacks-chow/tags.txt, source lines 11503–11503. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let U be a quasi-compact object of $\mathcal{C}$. Then the functor $\mathrm{Hom}_{\mathcal{O}}(j_!\mathcal{O}_U, -)$ commutes with direct sums.*

Proof. This is true because $\mathrm{Hom}_{\mathcal{O}}(j_!\mathcal{O}_U, \mathcal{F}) = \mathcal{F}(U)$ by (0G1V) and because the functor $\mathcal{F} \mapsto \mathcal{F}(U)$ commutes with direct sums by Lemma 0935. □

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